Practice 09: Linear Regression

Task 1.

In the lecture slides on page 45, change the alpha value and perform training, then plot a graph similar to the one on page 67.

강의자료 45페이지에서

alpha를 변경해서 학습을 수행하고 67페이지와 같은 그래프를 그려라

Task 2.

Modify the algorithm so that training stops when the cost decrease halts.

When changing the alpha value, calculate the number of iterations (theta updates) needed for training and display it in a table or graph.

Cost 감소가 멈추는 시점에서 학습을 중단하게끔 알고리즘을 수정하여라.

alpha값을 변경할 때, 학습에 소요된 iteration의 수(theta 업데이트 횟수)를 계산하여

표 혹은 그래프로 나타내어라.

Task 3.

Complete the <fill in> section on page 65 of the lecture slides and verify that the training proceeds correctly. Generate a graph similar to the one on page 67.

강의자료 65페이지의 <fill in> 부분을 완성하고 정상적으로 학습이 되는 것을 확인하여라.

67페이지와 같은 그래프를 완성하여라.

Task 4.

Using your implemented linear regression code, train a model to predict real estate prices (using the (**Real estate valuation data set.xlsx**). file). The given variables are as follows:

Data description:

Real estate valuation data collected from Sindian Dist., New Taipei City, Taiwan.

Input variables:

X1: transaction date (e.g., 2013.250 = March 2013, 2013.500 = June 2013, etc.)

X2: house age (unit: years)

X3: distance to the nearest MRT station (unit: meters)

X4: number of convenience stores within walking distance (integer)

X5: geographic coordinate, latitude (unit: degrees)

X6: geographic coordinate, longitude (unit: degrees)

Output variable:

Y: house price per unit area (unit: 10,000 New Taiwan Dollar/Ping, where 1 Ping ≈ 3.3 square meters)

Among these, use X2, X3, and X4 as input variables and Y as the target variable.

직접 구현한 linear regression 코드를 사용해서 부동산 가격 예측하는 linear regression model을 학습하여라 **(Real estate valuation data set.xlsx)**. 주어진 변수는 아래와 같다.

The market historical data set of real estate valuation are collected from Sindian Dist., New Taipei City, Taiwan.

The inputs are as follows

X1=the transaction date (for example, 2013.250=2013 March, 2013.500=2013 June, etc.)

X2=the house age (unit: year)

X3=the distance to the nearest MRT station (unit: meter)

X4=the number of convenience stores in the living circle on foot (integer)

X5=the geographic coordinate, latitude. (unit: degree)

X6=the geographic coordinate, longitude. (unit: degree)

The output is as follow

Y= house price of unit area (10000 New Taiwan Dollar/Ping, where Ping is a local unit, 1 Ping = 3.3 meter squared)

이 중, 입력변수는 X2, X3, X4 만 사용하고 목적변수는 Y를 사용한다.

Task 5.

Compare the linear regression model trained using the lm function with the model obtained in Task 4. Assess and compare the performance of the trained models.

lm함수를 사용해서 학습한 linear regression 모델과 Task 4에서 얻은 linear regression 모델을 비교해보자. 학습한 모델을 비교해보고, 모델의 성능도 비교해보자.